The attention mechanism can be plainly written as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

The self-attention mechanism can be written as:

$$\begin{aligned}&\text{Self Attention}(X; W_q, W_k, W_v)\\ &\quad= \text{softmax}\left(\frac{(XW_q)(XW_k)^\top}{\sqrt{d}}\right)(XW_v)\end{aligned}$$

The cross-attention mechanism can be written as:

$$\begin{aligned}&\text{Cross Attention}(X_q, X_c; W_q, W_k, W_v)\\ &\quad= \text{softmax}\left(\frac{(X_qW_q)(X_cW_k)^\top}{\sqrt{d}}\right)(X_cW_v)\end{aligned}$$

The relational-attention mechanism can be written as:

$$\begin{aligned}&\text{Relational Attention}(X_q, X_c; W_q, W_k, V)\\ &\quad= \text{softmax}\left(\frac{(X_qW_q)(X_cW_k)^\top}{\sqrt{d}}\right)V\end{aligned}$$

The adaptable-attention mechanism can be written as:

$$\begin{aligned}&\text{Adaptable Attention}(X_q, X_c; \gamma, W_q, W_k, W_v)\\ &\quad= \text{softmax}\left(\frac{(X_qW_q)(X_cW_k)^\top}{\sqrt{d}}\right)[(1-\gamma)X_c + \gamma I]\,W_v\end{aligned}$$

The pairwise transitive inference task can be described by:

Obs.?	x_i	x_j	$x_i > x_j$
Yes	A	B	1
	B	C	1
	B	A	0
	C	B	0
No	A	C	1
	C	A	0

Table 1: Truth table for transitive inference.

Context Identification Task

The context identification task requires identifying which context item is closest (in feature space) to a reference query.

Inputs (cross-attention mode):

- $X_q \in \{0, 1\}^{1\times L}$: Query (reference) one-hot vector

- $X_c \in \{0,1\}^{L \times L}$: Context matrix of one-hot vectors

Example ($L = 5$): Reference has feature index 2.

$$X_q = \begin{bmatrix} 0 & 0 & \mathbf{1} & 0 & 0 \end{bmatrix} \quad \text{(reference: feature 2)}$$

$$X_c = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{array}{l} \leftarrow \text{feature 4, dist=2} \\ \leftarrow \text{feature 0, dist=2} \\ \leftarrow \text{feature 0, dist=2} \\ \leftarrow \text{feature 1, dist=1 (closest)} \\ \leftarrow \text{feature 4, dist=2} \end{array}$$

Targets:

Feature target (which *feature index* is closest):

$$y_{\text{feature}} = \begin{bmatrix} 0 & \mathbf{1} & 0 & 0 & 0 \end{bmatrix} \quad \text{(feature index 1)}$$

Position target (which *context position* is closest):

$$y_{\text{position}} = \begin{bmatrix} 0 & 0 & 0 & \mathbf{1} & 0 \end{bmatrix} \quad \text{(context position 3)}$$